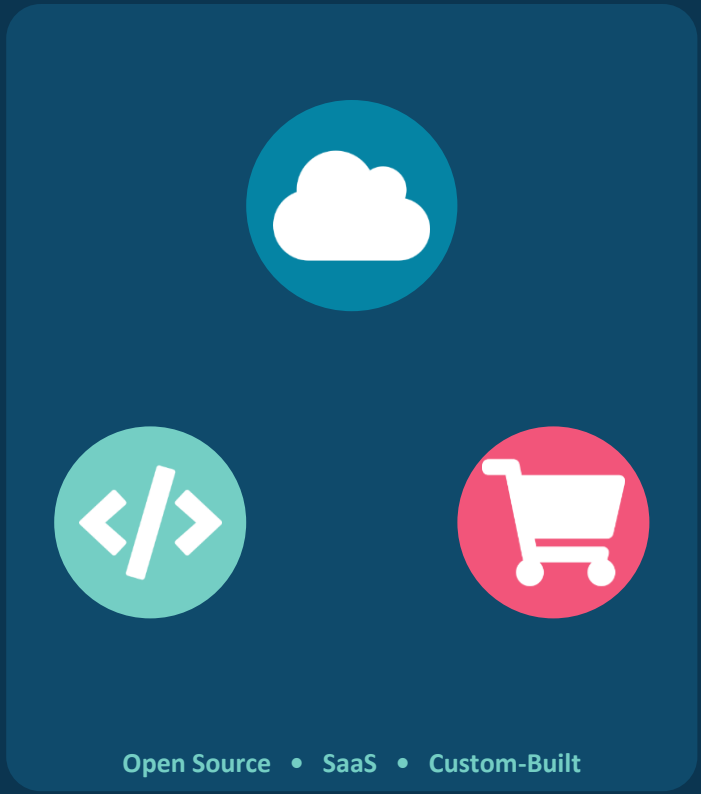
ICT2142 E-Business Systems

Lecture 06

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E-Commerce Platforms and Tools

Learning Objectives

- Identify the main categories of e-commerce platforms: open-source, SaaS and custom-developed solutions.
- Explain the key features, benefits and limitations of each platform type.
- Describe situations where a custom-built solution is more appropriate than an off-the-shelf platform.
- Apply practical criteria — cost, scalability, technical skill and business needs — to select the right e-commerce platform.

Open-Source E-Commerce Platforms

Open-source platforms provide free, publicly available source code that a business can download, host and customize itself. Examples include WooCommerce, Magento Open Source, PrestaShop and OpenCart.

- Free to use, but hosting, development and maintenance still cost money
- Full control over source code, design and functionality
- Requires in-house or hired technical expertise to build and maintain
- Large plugin/extension ecosystems and community support
- The business is responsible for security updates and scaling



Best suited for:

Businesses with technical skills in-house that want complete ownership and customization.

SaaS E-Commerce Platforms



Best suited for:

Startups and SMEs that want to launch quickly with minimal technical overhead.

Software-as-a-Service (SaaS) platforms are fully hosted and managed by a third-party provider on a subscription basis — for example Shopify, BigCommerce and Wix eCommerce.

- Quick to set up — no servers or infrastructure to manage
- Provider handles hosting, security, updates and scalability
- Monthly/annual subscription fee; customization is more limited
- Built-in payment gateways, themes and an apps/add-ons marketplace
- Minimal technical expertise needed to launch and run the store

Custom-Developed E-Commerce Solutions

A custom-developed solution is built from scratch (or heavily customized) using web frameworks and original code, so it can precisely match a business's unique requirements and existing systems.

- Complete flexibility — tailored features, workflows and UX
- Deep integration with existing ERP, CRM or legacy systems
- Higher development cost and longer time-to-market
- Full ownership of code and data, with no licensing limits
- Needs a dedicated development and maintenance team



Best suited for:

Large enterprises with unique, complex requirements that no existing platform can meet.

Comparing the Three Approaches



Open-Source

- Low license cost, DIY hosting
- High customization
- Needs technical skill
- You manage security/updates



SaaS

- Subscription fee, hosted for you
- Limited customization
- Little technical skill needed
- Provider manages security/updates



Custom-Built

- Highest cost & timeline
- Unlimited customization
- Requires dev team
- Full control & ownership

Selecting the Right Platform

Key Decision Factors



Budget

Upfront cost vs. ongoing subscription fees.



Technical Expertise

Skills available in-house or via hire.



Customization Needs

How unique the required features are.



Time-to-Market

How quickly the store must launch.



Scalability

Expected growth in traffic and orders.



Integration Needs

Connecting with ERP, CRM or other systems.

Summary & Key Takeaways

- Open-source platforms give full control and customization, at the cost of needing in-house technical skill.
- SaaS platforms let a business launch quickly with low technical overhead, in exchange for less flexibility.
- Custom-developed solutions offer complete flexibility for unique needs, but require the biggest investment.
- The right choice depends on budget, technical expertise, customization needs, timeline, scalability and integration requirements — not on which platform is “best” in general.

Sample Questions

1. Briefly explain why a startup with limited technical staff and a need to launch quickly would typically prefer a SaaS platform over a custom-developed solution.

2. List and briefly describe THREE factors a business should consider when choosing between an open-source, SaaS, or custom-developed e-commerce platform.

Thank you… !